

Yash Chauhan

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EDUCATION

SRM Institute of Science and Technology, Kattankulathur, Chennai Aug 2023 - Expected 2027
B.Tech in Computer Science and Engineering, Specialization in Artificial Intelligence and Machine Learning **CGPA: 8.5/10**
SRM Nagar, Kattankulathur - 603203, Tamil Nadu, India

- **Competitions:** IIS2 Hack2Skill 2024 - top 35% among 300+ teams; Hack 3.0 Street participant.
- **Activities:** GitHub Community SRM - Technical Associate supporting open-source contributions; Indian Society for Technical Education - Student Member.

Sanjay Ghodawat International School, Kolhapur, Maharashtra 2023
Class XII, CBSE - Physics, Chemistry, Mathematics and Biology **70%**

Sanjay Ghodawat International School, Kolhapur, Maharashtra 2021
Class X, CBSE - English Language & Literature, Mathematics Standard, Science, Social Science, Hindi Course-B **89%**

WORK EXPERIENCE

Co-Founder & Founding Engineer, Gridee | [Google Play](#) | [App Store](#) Jan 2026 - Present
Technologies: Kotlin, CameraX, ML Kit OCR, ZXing, Retrofit, OkHttp, JWT, Razorpay, React, TypeScript, Vite

- Co-founded and shipped an end-to-end smart-parking platform with live availability, advance and instant reservations, wallet/Razorpay payments, and **8,000+ combined Android and iOS downloads**.
- Implemented QR and number-plate check-in/out using CameraX, ML Kit OCR, and ZXing, with operator workflows for vehicle entry, exit, and occupancy management.
- Secured mobile-to-API communication with JWT, Retrofit, and OkHttp while supporting coin-based refundable bookings and cashless access flows.
- Added email/password and Google sign-in, user-facing privacy controls, and Razorpay checkout without storing card details.

Research Intern, Indian Institute of Technology Madras, Chennai Mar 2025 - Mar 2026
Technologies: Python, Pandas, NumPy, SciPy, scikit-learn, XGBoost, SHAP, GraphSAGE, variational autoencoders, OSMnx, NetworkX, Folium, Docker

- Automated ingestion and validation of **5+ years of ambulance-dispatch and road-construction data** with **100+ quality checks**; simulated **5,000+ routes** under peak-hour, monsoon, and construction-delay scenarios.
- Engineered a **55-column analytical matrix** and weighted GraphSAGE-VAE-XGBoost workflow for stage-wise representation learning, anomaly detection, contextual prediction, and operational-drift analysis.
- Built an earlier two-phase benchmark in which an autoencoder identified **8,567 outliers** and informed a Time Drift Index; an XGBoost counterfactual model achieved **8.37% WMAPE**.
- Designed a fairness-aware Quantum Stability Index normalized against call volume, classifying districts into **five interpretable stability bands** for policy targeting.
- Applied t-tests, ANOVA, Kruskal-Wallis tests, power analysis, XGBoost, and SHAP to distinguish statistical significance from operational relevance and identify key delay drivers.
- Conducted equity analysis across metro, tribal, rural, APL, pediatric, and behavioral-case contexts to support resource allocation and accountability decisions.
- Produced interactive geospatial dashboards and publication-quality visualizations for response-time hotspots, route constraints, and stakeholder review.

Summer Intern - Electrical and Instrumentation, Hindalco Industries Limited, Mumbai, Maharashtra Jun 2025 - Jul 2025
Technologies: MTell, Proficy Historian / OSI PI, MES, Python, ARIMA, custom anomaly detection, Power BI

- Extracted and transformed high-frequency sensor streams from Proficy Historian through MES pipelines for asset-level reliability analysis.
- Built predictive-maintenance models in MTell using ARIMA and custom anomaly-detection techniques.
- Integrated real-time anomaly alerts into maintenance workflows to enable proactive scheduling and optimized service intervals.
- Helped deliver reported reductions of **20% in unscheduled downtime** and **15% in maintenance cost** through earlier failure detection.

RESEARCH & PUBLICATIONS

Digital Inclusion and Circular Economy Behavior in Indian States:

A Digital Circularity Index and Explainable XGBoost Analysis

First Author - Submitted and under double-blind review, 5th IEEE Delhi Section International Conference (IEEE DELCON 2026)

- Integrated seven official government datasets across 16 states to construct a Digital Circularity Index; validated the digital-access relationship using OLS ($R^2=0.610$, $p<0.001$), permutation tests, rank statistics, and bootstrap checks.
- Built leak-free nested LOO-CV XGBoost ($R^2=0.511$), SHAP explanations, and K-Means state typologies revealing a **2.5x DCI gap** between Digital Leaders and Emerging States.

Performance at Scale: Detecting Operational Drift and Structural Inequity in Tamil Nadu's Emergency Medical Services

Third author (of three) - Manuscript submitted to Transportation Research Part C: Emerging Technologies

- Analyzed **3.59 million** Tamil Nadu 108 dispatch records from **2017-2025** across 38 districts, 44 emergency types, five operational legs, and seven equity dimensions.
- Developed a GraphSAGE-VAE-XGBoost framework and Composite Burden Index to separate operational drift from contextual constraints and surface actionable inequities.

TECHNICAL PROJECTS

AutoScaler - AI-Driven Kubernetes Observability and Self-Healing System

Mar 2026 - Present | [GitHub](#)

Technologies: Python, FastAPI, TensorFlow/Keras, scikit-learn, SHAP, NestJS, Next.js, TypeScript, Recharts, D3.js, Prometheus, TimescaleDB, Redis Pub/Sub, Docker, Kubernetes, RBAC, Socket.IO, Jaeger

- Deployed a three-node Kind cluster with Prometheus discovery, TimescaleDB persistence, Redis coordination, analysis services, and healing controllers running as isolated Kubernetes workloads with RBAC.
- Implemented statistical baselines and Isolation Forest detection with live SHAP feature-importance charts; designed a documented **7-day cold-start warm-up** before telemetry-gated LSTM activation.
- Built automatic Claude-to-OpenAI-to-in-cluster Random Forest failover that preserves an **identical JSON output structure**; the fallback path runs with **zero external dependency**.
- Built a Chaos Engine injecting six failure types across three intensity levels on a **randomized schedule**, paired with a watchdog checking seven core components every 30 seconds.
- Delivered nine routed Next.js views - seven live plus telemetry-gated SLO and Settings views - and instrumented a pending **48-hour SLO validation** for detection rate, recovery time, and false positives.

Pulse - AI-Native Campaign Platform

[Live](#) | [GitHub](#)

Technologies: TypeScript, Next.js 16, NestJS, Express, Prisma, PostgreSQL, BullMQ, Redis, Zod, REST APIs

- Built and deployed a three-service monorepo with shared Zod contracts, editable audience rules, match-count previews, and campaign workflows across WhatsApp, SMS, email, and RCS.
- Compiled natural-language targeting requests into a validated Segment DSL and parameterized Prisma queries, isolating the LLM from SQL, customer PII, and direct database access.
- Engineered rate limits, HMAC callbacks, idempotent events, retries, dead-letter queues, and automatic channel failover.
- Added conversion attribution, channel and revenue breakdowns, and on-read funnel analytics over **5,000 seeded customers and 25,000 orders**.

Denoising and Quality Enhancement of CT Scan/X-Ray Images

Feb 2025

Technologies: Python, TensorFlow, Keras, CUDA/cuDNN, NumPy, Pandas, OpenCV, scikit-image, scikit-learn, tqdm, Matplotlib, Seaborn, Jupyter

- Engineered a mixed-precision U-Net with an approximately **12 dB PSNR gain** and **95% noise reduction**; 95% of 300+ validation scans improved jointly across PSNR, SSIM, and MSE.
- Built a memory-efficient generator for dynamic noisy-clean pairing and accelerated convergence approximately **4x** through mixed precision, adaptive learning rates, and early stopping.
- Automated loss curves, side-by-side comparison panels, and per-image evaluation dashboards; documented curation, CLAHE preprocessing, augmentation, and future GAN/diffusion work.

GlassBox - Auditable Wealth-Advisory AI Agent

Feb 2026 - May 2026 | [GitHub](#)

Technologies: Python, FastAPI, Uvicorn, SQLAlchemy, Alembic, PostgreSQL, ChromaDB, sentence-transformers, scikit-learn, Next.js, React, TypeScript, Recharts, Docker, Playwright, pytest

- Built an auditable advisory and compliance assistant with grounded citations, claim verification, refusal handling, deterministic evaluation, and replayable audit trails.
- Delivered FastAPI, Next.js, PostgreSQL, and ChromaDB workflows with automated migrations, tests, containerized services, a live trust dashboard, and decision replay.
- Trained scikit-learn grounding, refusal-routing, and fallback classifiers; surfaced hallucination, determinism, refusal, and audit-completeness metrics in Recharts.

Kalakraft - Full-Stack Marketplace for Hand-Crafted Resin Art

Jun 2025 - Sep 2025 | [Live](#)

Technologies: Next.js, React, TypeScript, Tailwind CSS, Node.js, PostgreSQL, Prisma, Firebase, NextAuth.js, Cloudinary, Pusher, Sendinblue, Vercel

- Created a production storefront and admin dashboard covering purchases, order tracking, inventory, coupons, reviews, users, support tickets, and revenue metrics.
- Built a Sharp-Cloudinary media pipeline and secure role-based, multi-provider authentication using Firebase and NextAuth.js.
- Implemented Pusher order updates and low-stock alerts, plus Sendinblue flows for **order confirmations, password resets, and coupon campaigns**; automated Vercel deployments.

SKILLS

Programming Languages: Python, C++, SQL, TypeScript, Kotlin

ML, Deep Learning and Computer Vision: TensorFlow, PyTorch, Keras, scikit-learn, XGBoost, SHAP, GraphSAGE, VAEs, anomaly detection, medical imaging, OpenCV, ML Kit OCR

Data and Analytics: Pandas, NumPy, SciPy, tqdm, statistical testing, power analysis, ETL pipelines, SQL database management, Power BI, Matplotlib, Seaborn, Folium, JupyterLab

Backend, Web and Mobile: FastAPI, Uvicorn, Node.js, NestJS, Express, Next.js, React, React Native, Recharts, D3.js, Tailwind CSS, Firebase, NextAuth.js, Razorpay, REST APIs, WebSockets, User Authentication, Role-Based Access Control

DevOps and Infrastructure: Docker, Kubernetes/Kind, RBAC, Prometheus, TimescaleDB, PostgreSQL, Prisma, Redis Pub/Sub, BullMQ, Git, CI/CD, Jaeger

Programming Fundamentals: Object-oriented programming, data structures and algorithms, operating systems, computer organization and architecture

CERTIFICATIONS

- [AWS Certified Cloud Practitioner](#) (Jan 2026 - Jan 2029) | [AWS Certified Machine Learning - Specialty](#)
- [Oracle Database SQL Certified Specialist](#) (May 2026) | [Salesforce Certified Agentforce Specialist](#) (Dec 2025; Credential ID: 7238407)
- [Deep Learning Specialization](#) (Aug 2023) | [Machine Learning Specialization](#) (Jun 2024) | [Mathematics for Machine Learning and Data Science](#) (Jul 2025)
- MathWorks: [Machine Learning with MATLAB](#), [MATLAB Onramp](#), [Machine Learning Onramp](#), and [MATLAB for Data Processing and Visualization](#) (May 2024)